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Emulate or differentiate?

Chinese development finance, competition, and World Bank infrastructure funding

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Abstract
Foreign aid relationships are valuable to donors as a means of improving development outcomes and influencing recipient country policy. The emergence of new donors can lead to competition as donors vie for influence over recipient government policy and attention. How does such competition affect the behavior of traditional donors? I draw attention to how the rise of China as a provider of development finance is changing the *type* of development that traditional donors support. Chinese development finance is particularly targeted at large infrastructure projects, and this focus can exert pressure on traditional donors. I suggest traditional donors can either emulate China's approach to development, i.e. offer projects in infrastructure-intensive sectors, or differentiate themselves and specialize in alternative approaches to development, e.g. focus on governance and social sector interventions. I test this using data on the terms of World Bank and Chinese development finance in over 100 countries. I find the World Bank responds to competitive pressure from China by emulating the Chinese emphasis on infrastructure, allocating a greater share of its development projects in infrastructure-intensive sectors when recipient countries receive more Chinese development finance. Furthermore, subnational data shows that the World Bank also emulates China's approach to development in response to competition at the regional level. China's growing role as a provider of development finance affects traditional donor behavior, shaping the type of development donors support by introducing bottom-up competitive pressure.

Keywords Development finance · Aid · China · World bank · Competition · International political economy · Infrastructure

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Since the end of the Cold War, advanced economies have been largely alone in providing development finance to developing countries, whether bilaterally or through multilateral institutions. For donor countries, these relationships with recipient governments offer a means of influencing recipient country policies, including to improve development outcomes in recipient countries. However, the rise of emerging donors, especially China, has disrupted the virtual monopoly previously enjoyed by traditional donors. Since recipient governments have a finite capacity to absorb external finance, alternative sources of development finance can place competitive pressure on traditional donors. I investigate how traditional donors respond to the arrival of China as a donor, contributing to a growing literature on interaction and competition among donors as the landscape of development finance becomes more crowded (Hernandez 2017; Swedlund 2017b; Granath 2016). I focus on the competitive pressure arising from the focus of Chinese development finance on large infrastructure-intensive projects. I suggest traditional donors, especially the World Bank, can respond to China's growing role by either emulating China's approach or differentiating the development assistance they offer.

Consider an example of emulation. In 2015, the World Bank announced it would provide a $700 million guarantee to help construct the Sankofa natural gas plant in Ghana. The guarantee was the largest ever of its kind from the World Bank (2015). China's growing role as a donor in Ghana was likely a factor in the Bank's decision to provide this unprecedented guarantee. Only a few years earlier, the Ghanaian government snubbed the World Bank by choosing to fund a different gas project with a loan from a Chinese creditor, despite initially planning the project with the World Bank (Boadu and Aklorbortu 2015). The Ghanaian press and donor circles speculated that the government considered the World Bank's environmental safeguard requirements too onerous and instead chose Chinese financing for its speed and ease. That the World Bank subsequently provided an unprecedented guarantee for a similar project is an example of how competitive pressure from China can lead traditional donors to *emulate* China.

However, emulation of China's approach is not the only possible response to competitive pressure. Donors may also choose to *differentiate* their approach. For example, when Ethiopia and the EU negotiated the focus of development cooperation for 2014-2020, the EU announced its intention to withdraw from the road sector. Though noting that "the road sector [in Ethiopia] remains one of least developed in Sub-Saharan African countries," the EU decided to shift its financing out of the roads sector and into other sectors, especially social sectors, including the health sector (Government of Ethiopia and European Commission 2014: 14). This shift occurred against the backdrop of a growing Chinese presence in the Ethiopian road sector, including a $320 million loan in 2013 from the China Export Import Bank to finance a ring road around the capital Addis Ababa.[1] When confronted with competition from China, traditional donors may also choose to *specialize* in alternative sectors.

[1] By comparison, the EU had previously allocated roughly EUR 280 million to the *entire* road sector in Ethiopia over a period of 6 years, 2008-2013.

China's infrastructure-led approach to its own development has received considerable attention, as have its efforts to export this approach abroad by providing funding for similar infrastructure investments, including through the One Belt One Road initiative (Ansar et al. 2016; Fukuyama 2016). In addition, China may have an effect on the activities of traditional donors through the competition it introduces at the recipient country level, spurring traditional donors to shift their funding to different kinds of projects. Previous work on the consequences of new donors has focused on whether donors reduce the conditionality of their aid in response to Chinese competition (Hernandez 2017). Building on this work, I draw attention to other distinctive attributes of Chinese financing, namely the focus on infrastructure-intensive sectors.

Traditional donors could respond to this competition from China's development orientation in a number of ways, including emulation or differentiation. Traditional donors could imitate the Chinese approach to development finance, shifting their development projects into infrastructure-intensive sectors. Alternatively, they could differentiate themselves from the Chinese approach, focusing on social sector projects in health and education that are complementary to Chinese finance. There are incentives for both possible responses, and it is not clear ex ante which is the most likely traditional donor response to competition, nor whether traditional donors will feel pressured to respond to competition in the first place.

I study these dynamics with respect to the World Bank, the world's leading multilateral donor. For the World Bank, competition with China arises out of reduced access to recipient governments to shape policy outcomes, as well as internal pressure to extend loans to meet lending targets. Furthermore, the strategic considerations of shareholder countries vis-à-vis China may enter the World Bank's decision-making process.

To capture how the World Bank responds to competition from China, I analyze the share of annual World Bank projects in recipient countries that are in infrastructure-intensive sectors, including energy, transport and industry, over the period 2000-2014. The key explanatory variable is Chinese official financing commitments, taken from the AidData Global Chinese Official Finance dataset (Dreher et al. 2017). To address concerns about the endogeneity of Chinese finance, I employ an instrumental variable approach that uses Chinese steel production to instrument for Chinese overseas finance. As an important input into projects funded by Chinese overseas finance, excess steel production provides an exogenous shock that, when interacted with a country's probability of receiving Chinese finance, yields an instrument that varies over time and across countries. The instrument has been employed in a number of other studies of Chinese official finance (Dreher et al. 2017, 2019; Bluhm et al. 2018) and allows me to estimate the causal effect of Chinese finance commitments on traditional donor behavior.

I find consistent evidence that the World Bank responds to competition by emulating the Chinese approach to development finance. When recipient countries receive more Chinese development finance, the World Bank subsequently allocates a greater share of its development projects to infrastructure-intensive sectors such as transport or energy. This same pattern holds at the subnational level. When regions receive greater amounts of Chinese development finance, the World Bank subsequently

emulates a Chinese approach to development finance, allocating a greater share of its projects to infrastructure-intensive sectors in that region.

The paper proceeds as follows. Section 1 describes the key features of Chinese development finance, comparing it to traditional donor financing to outline how the differences in development finance are a source of competitive pressure. Section 2 sets out the core argument that traditional donors are likely to perceive China's growing role as a donor as a form of competition and that strategies in response to this competition include emulation and differentiation of the Chinese approach. Section 3 discusses the data and the methods employed, Section 4 presents and analyses the results, and Section 5 offers a brief conclusion.

1 Introducing competitive pressure? Characteristics of Chinese development finance

As a relatively new development finance actor, China has taken an approach that diverges from the practices of traditional donors. Existing work on China's international development practices has, for instance, focused on the fact that Chinese aid and loans are not accompanied by the governance and performance conditionality that accompany traditional donors' development finance (Hernandez 2017; Swedlund 2017b). I identify additional distinctive features of Chinese finance that enhance its attraction for recipient countries, namely the sectoral focus of Chinese-funded projects, which are more likely to be targeted at infrastructure-intensive "productive" sectors.

China has substantially increased its official finance to developing countries over the past decade (Fig. 1).[2] Data from AidData's Global Chinese Official Finance dataset shows the steady growth of Chinese overseas financing commitments from a low base in the early 2000s. China committed $2.6 billion in development finance in 2000, an annual commitment that expanded to over $36 billion by 2013.[3] Over the past decade, Chinese overseas finance commitments have caught up to the World Bank's global portfolio of development projects, even surpassing the Bank in some recent years.

[2]Note that in order to encompass both the concessional and non-concessional finance provided by China I refer to development finance or overseas finance rather than "aid," since aid or overseas development assistance (ODA) has clear requirements with respect to concessionality. As defined by the OECD, in order to qualify as ODA, a project must have development as its purpose and at least a 25% grant element. Much of China's development finance does not meet this definition.

[3]In both the descriptive statistics that follow and the analysis in Section 4, I use data on donors' financing commitments, rather than disbursements. In part, this is due to data availability. The AidData Global Chinese Official Finance dataset uses the "Tracking Underreported Financial Flows" (TUFF) methodology, which provides data only on commitments, not disbursements of Chinese finance. For direct comparison, I therefore use data on World Bank commitments, rather than disbursements, though the amount disbursed can differ from the amount committed. While studies of the effects of aid and finance require data on the amount of finance actually received, commitments data is suitable for studying the *intentions* of donors and therefore appropriate for studying competition. Disbursements can sometimes vary for reasons that are out of the control of donors, while commitments are fully determined by donors.

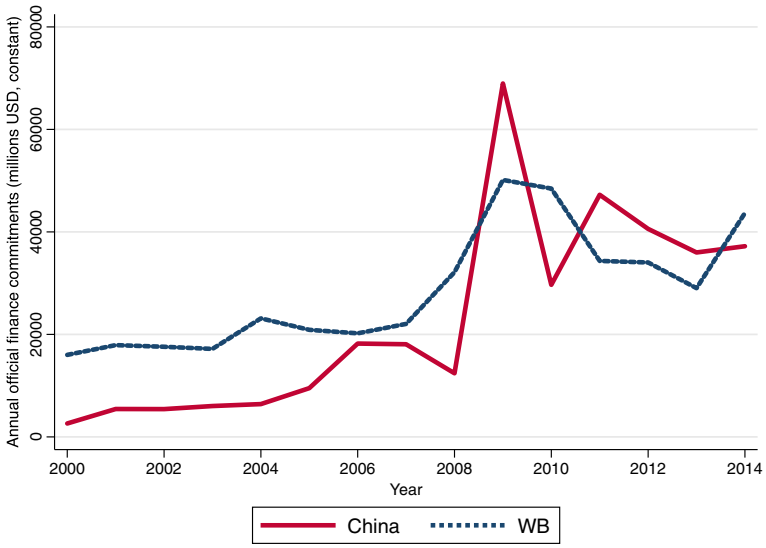

Fig. 1 Annual official finance commitments: China and the World Bank, 2000-2014

A considerable body of scholarship has emerged to explain the expansion of Chinese development finance. Most trace the growth in Chinese overseas financing to China's growing global ambitions, an accumulation of dollar reserves arising from China's significant trade surplus and a desire to support Chinese firms expanding abroad (Alden 2007; Brautigam 2011; Dollar 2018; Taylor 2011). Critics often assert that Chinese aid is more commercially and politically motivated than that of traditional donors, particularly that it is targeted towards countries rich in natural resources, politically aligned with China, or governed by autocratic rulers (Rotberg 2008; Naím 2009). However, Dreher and Fuchs (2015) find no effect of natural resource wealth or democracy on the allocation of Chinese concessional funds that most closely resemble traditional donors' aid. Similarly, Dreher et al. (2018) stress that Chinese concessional commitments are not motivated by natural resource wealth or decisively shaped by regime type and institutional quality, though less concessional forms of official finance do appear to be driven by commercial considerations, including natural resource wealth.

One notable difference between Chinese development finance and traditional donor aid is the lack of policy conditionality attached to Chinese projects. While traditional donors often attach reform or policy requirements to their development assistance, frequently as a precondition for releasing aid, China has emphasized "non-interference" and mutual respect for sovereignty as central to its development program, and has openly distanced itself from the conditionality approach employed by traditional donors (People's Republic of China 2011; Six 2009; Alden 2007). Though Chinese official finance often comes in the form of export credits from China's ExIm bank, and thus requires the project in question be carried out by a Chinese firm, there are no policy or reform conditions beyond these commercial

requirements. Hernandez (2017) has drawn attention to how this difference between Chinese development finance and traditional donor aid might place pressure on traditional donors, leading them to reduce the conditionality of their aid. He finds that the World Bank attaches fewer conditions to its loans when a recipient country receives greater commitments of Chinese finance.

Chinese development finance is distinctive from traditional donor aid in ways other than the lack of policy conditionality. I argue these other distinctive characteristics also have the potential to disrupt traditional donor behavior. Namely, Chinese development projects tend, on average, to be large and targeted towards infrastructure projects, especially in energy and transport sectors. Though these differences between Chinese development finance and traditional donor aid have been recognized in the literature, their effect on traditional donors has yet to be empirically investigated.[4]

These differences between Chinese development finance and traditional donor aid are attributable to a number of factors. First, since export credits make up a large part of Chinese official finance, there is a tendency for projects to draw on Chinese companies' competitiveness in construction (Brautigam 2011). Second, since a considerable portion of Chinese financing is non-concessional, projects must have the potential to earn returns that will allow repayment. Therefore, there is a bias towards projects with a clear economic return, leading to a greater emphasis on infrastructure and productive sectors over health or education.

Furthermore, China stresses that its development finance is "demand-led" and responds to recipient countries' appetite for infrastructure projects. This combines with China's emphasis on infrastructure-led growth in its *own* development. The Chinese discourse on its development assistance stresses the potential for infrastructure projects to have a transformative effect on the economy (Kragelund 2014). This differs from the approach of many of the traditional donors, whose portfolio tended towards social sectors and smaller projects over the course of the 2000s, as aid budget shrank and the Millennium Development Goals emphasized achieving social sector objectives (Swedlund 2017a: 9). While this trend was less pronounced for the World Bank and other multilateral development banks, they too shifted funds into social sector programs in education and health.[5] In particular, the increased sensitivity to environmental and social impacts of large-scale infrastructure projects has meant that funding such projects with the World Bank now requires extensive and lengthy environmental and social impact reviews (Park 2007). By contrast, Chinese funders are reputed to move much more quickly, without formalized and extensive environmental review procedures.

[4]By drawing attention to these differences in the sectoral allocation of different donors' development finance, I follow the advice of Nielson et al. (2017), who encourage researchers to further disaggregate analyses of development finance by sector and modality.

[5]In fact, the World Bank's perceived neglect of infrastructure projects and productive sectors has been a long-standing concern for some recipient countries, as Kellerman (2019) describes with respect to Asian countries' criticisms of the World Bank in the 1960s.

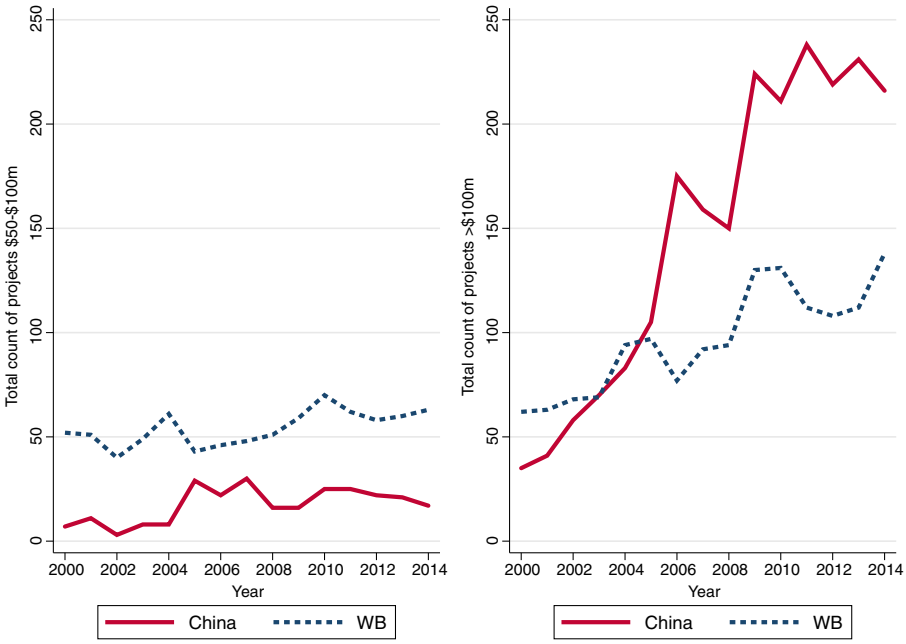

Fig. 2 Global count of projects 50-100 million USD and over 100 million USD: China and the World Bank, 2000–2014

These differences between Chinese and traditional donors' approaches to development projects are reflected in the data. Figure 2 shows the number of large projects funded annually by China and the World Bank. The Chinese emphasis on "megaprojects" distinguishes its development finance from that of traditional donors. While the average World Bank project between 1999 and 2015 cost $76.96 million, the average Chinese-financed project during this time cost $132.21 million.

Importantly for recipient governments, Chinese-financed projects are also distinctive in their sector focus. Figure 3 shows the proportion of annual projects donors allocated to projects in sectors such as the energy, transport or industry, where infrastructure projects are more likely.[6] A greater percentage of Chinese-financed projects are in these sectors, though the share of World Bank-funded projects in infrastructure-intensive sectors increased in the late 2000s.[7] Over the 2000-2014 period, 13.26% of World Bank-funded projects were in infrastructure-intensive sectors, while 27.44% of Chinese-financed projects were in the same sectors, a difference that is statistically significant.

[6]Projects are recorded as being in hard sectors if they fall into one of the following sectors in the OECD's Creditor Reporting System (CRS) sector classification scheme: (1) water supply and sanitation, (2) transport and storage, (3) communications, (4) energy, (5) agriculture, forestry & fishing, and (6) industry, mining, and construction.

[7]Humphrey notes that the increase in infrastructure investment in the late 2000s is part of counter-cyclical lending by multilateral development banks in the period after the Global Financial Crisis. See Humphrey (2015: 3-4).

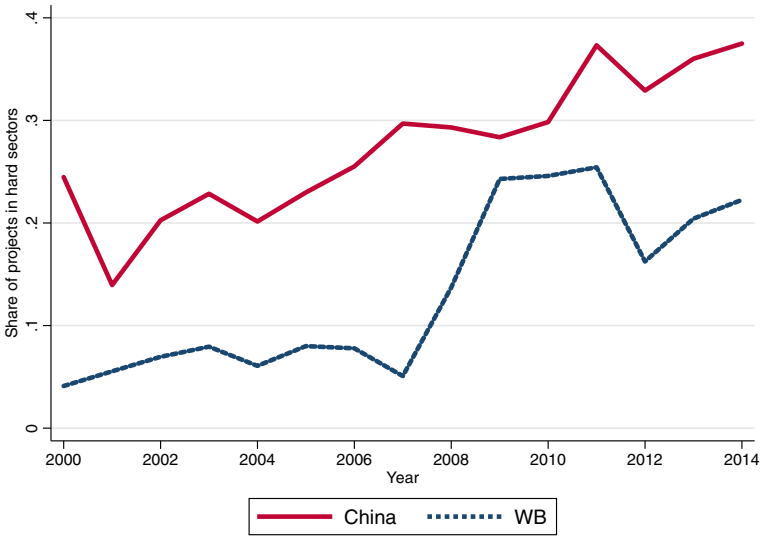

Fig. 3 Share of projects in "hard" sectors, i.e. energy, transport, or industry: China and the World Bank, 2000–2014

2 Competition and traditional donor strategies

The entrance of China as a new donor with different practices to those of traditional donors can introduce competition and disrupt the relationship between recipient governments and traditional donors. I suggest the entry of China has the potential to shape the *type* of development that traditional donors choose to support with their funding. If donors respond to the competition from Chinese development finance, they can do so by emulating the Chinese approach and funding projects in infrastructure-intensive sectors or differentiating themselves by specializing in alternative niches.

2.1 Competition among donors

The claim that the entrance of a new donor can lead to competition implies donors are contending for access to a scarce resource. Following others, I suggest there are two main reasons a traditional donor like the World Bank would compete with other donors. First, donors value their access to the recipient government, access that is (mostly) zero-sum. Second, donors are under pressure to disburse funds.

There is a considerable literature on donors' motivations in giving aid. See, for instance, Alesina and Dollar (2000), Dreher et al. (2013), Fuchs et al. (2014), Hoeffler and Outram (2011), Bermeo (2011, 2017). One motivation identified in this literature is donors' interest in influencing recipient policy (Bueno de Mesquita and Smith 2016). Donors may value their influence over recipient policy for self-interested reasons, such as securing recipient governments' support in international

organizations or favorable conditions for investors, or they may value their influence because they believe their research and experience across developing countries allow them to be effective in achieving development outcomes (Swedlund 2017a: 27). The latter is more relevant to multilateral donors like the World Bank, who seek access to recipients to translate their expertise into maximum development impact (Milner 2006: 109).

To have influence over government policy, donors need access to the recipient government, access they receive in connection with the development projects and programs they fund. In donor circles, this is referred to as "policy dialogue." Donors value their access to recipient governments and worry when exposure to senior policymakers in recipient countries is reduced. Competition arises when the recipient government allocates its time to meeting with other donors, thereby reducing the access traditional donors have to encourage the government to pursue their desired policies. Traditional donors have sought to mitigate the challenge of zero-sum access to the recipient government by coordinating their positions, such that one donor's access to the government does not necessarily take away from that of others.[8]

In the case of China, however, access to recipient governments is essentially zero-sum, since China does not participate in coordination and harmonization mechanisms with other donors (Bigsten and Tengstam 2015). While a recipient government's time spent with an OECD donor or the African Development Bank likely reinforces the World Bank's policy messages, this is not the case with China. When recipient governments dedicate time to meeting with China because they consider Chinese financing attractive, this reduces the access the World Bank has to engage the government on their desired policy outcomes.

A similar drive to compete is modeled by Bueno de Mesquita and Smith (2016), who focus on the competition arising from aid-for-policy deals. They assume donors provide aid in exchange for the recipient adopting policies desired by the donor. Competition arises when a rival donor offers aid in exchange for the recipient adopting a different policy. Since the recipient can only adopt one of these two policies, the outcome is necessarily zero-sum, and competition occurs. They find that donors are driven to provide increasing amounts of aid to secure their desired policy outcome. Using data spanning the periods before, during and after the Cold War period, they find that US aid allocation increases during periods of global competition with the Soviet Union.

It is not always the case that China and multilateral donors such as the World Bank pursue diverging objectives in recipient countries. In a number of cases China ExIm Bank and China Development Bank have co-financed programs with multilateral banks such as the African Development Bank. Furthermore, as a shareholder of the World Bank, China has some influence over World Bank policy. However, China's influence within the World Bank remains small compared to influential traditional donor shareholders. Despite recent capital increases and reallocating of voting shares

[8] In the 2005 Paris Declaration, donors in the OECD Development Assistance Committee (DAC) committed to "harmonizing" their aid programs. However, coordination has in practice not been as extensive as promised. See Aldasoro et al. (2009) and Nunnenkamp et al. (2013).

China's vote at the World Bank remains limited; 4.8% of the IBRD total vote share and 2.25% of the IDA total vote share, compared to 15.49% and 10.21% for the United States, for example.[9] Furthermore, there are significant divergences between the World Bank's approach to development finance and that of the major Chinese policy banks. Since China's policy banks largely do not coordinate their projects with the World Bank, the more time that recipients choose to dedicate to Chinese development projects, the fewer opportunities there are for the World Bank to pursue its own policy agenda with the recipient government. The constraint that induces competition is thus the access to the recipient government for policy influence.

In addition to the motivation to retain access to the government for "policy dialogue," there are bureaucratic drivers of competition for the World Bank. Donors compete for a share of the total development finance a country receives. As other work has highlighted, World Bank staff are under pressure to disburse funds.[10] Yanguas and Hulme (2015: 214) cite one World Bank practitioner who refers to the "overwhelming pressure to get money out of the door" and that "task teams are paid to ensure that the money is disbursed." Given this incentive to disburse, competition arises when recipients choose to accept development finance from other sources and then, for reasons of debt carrying capacity, reduce the amount they take from the World Bank.

This is most obvious with development finance that comes in the form of loans, such as those offered by the China ExIm Bank or the World Bank's International Bank for Reconstruction and Development (IBRD). Recipient governments are constrained in the amount of debt financing they can take on by the need to keep their total debt below a sustainable threshold.[11] With loan finance, competition arises because recipients can only borrow so much, and the choice of one lender therefore reduces the amount that another lender can lend. Combined with bureaucratic incentives to disburse, this can lead to competition among providers of development finance for the share of total external finance that a recipient takes on.

It may appear that such competition is less likely with grant aid and concessional finance, since recipients are not constrained by debt sustainability concerns in their decision of how much aid to receive. However, there are also limits to the total amount of grant aid a recipient can take on. These limits arise not from debt sustainability, but rather from the transaction costs associated with development projects and the absorptive capacity of the recipient's economy. Aid projects pose significant administrative costs for the recipient government, including the design of the project, monitoring of funds, and reporting to donors (Paul and Vandeninden 2012). In some cases, recipients are asked to provide a financial contribution, referred to as counterpart funding, towards the project funded with donor grants. Limits on the capacity and resources of the recipient will thus act as a constraint on the total amount of grant

[9] See http://pubdocs.worldbank.org/en/795101541106471736/IBRDCountryVotingTable.pdf and http://pubdocs.worldbank.org/en/845861541106477171/IDACountryVotingTable.pdf

[10] This is often discussed as an impediment to conditionality, since the Bank cannot credibly threaten to withhold funds in the case of non-compliance with conditions. See Svensson (2003)

[11] Bunte (2019) makes a similar point about the impact of debt constraints, highlighting that these constraints mean that a choice for one lender is a choice against another lender.

aid a country can take on.[12] These limits combine with the bureaucratic incentives to disburse and the desire for policy access to produce the potential for competition.

In sum, competition arises from the fact that donors seek access to the recipient government, access that is constrained by limits on the recipient government's time, policy space, and its ability to take on outside finance. Therefore, one can anticipate an effect of increasing Chinese finance on traditional donor behavior even though Chinese development finance and traditional donor aid tend to be quite different. In fact, it is precisely *because* Chinese development finance is different from traditional donor aid that Chinese finance places competitive pressure on traditional donors to change their own patterns of development finance.

These mechanisms underpinning competition largely play out at the level of the recipient government. An increase in Chinese development finance reduces the time and policy space that recipient governments can allocate to traditional donors, incentivizing the World Bank to offer more attractive development finance to ensure continued influence with the recipient government. However, these dynamics may also unfold at the subnational level, especially if donors value their access to municipal or local governments as implementers of development projects. Furthermore, competition may be localized if certain regions are perceived by donors to be more attractive for development projects, for instance because government capacity is higher. Therefore one might expect strategies of emulation or differentiation to also occur at the subnational level.

2.2 Recipient preferences

Having established donor motivations that can lead to competition, it is important to briefly specify recipient preferences with respect to development finance, since recipient preference for one form of donor financing rather than another underpins donors' strategies in response to competition. I suggest recipients have a preference for projects that are larger and targeted at infrastructure. These preferences arise from the recipient government's political incentives.

Recipients' preferences with respect to the *type* of development interventions supported by outside finance stem from political imperatives to deliver tangible and visible development achievements. This underpins the preference for infrastructure-intensive projects in productive sectors that are more readily communicated to constituents than more diffuse development interventions, for example in governance or in the health sector. Governments in most developing countries derive their legitimacy from delivering development gains, which are easiest to demonstrate with highly visible interventions, such as investments in public infrastructure. Foreign aid

[12]These constraints on recipient countries' absorptive capacity for grant-funded projects are one reason that budget support is particularly attractive for recipient countries, since budget support releases donor funds directly into the government's coffers. This suggests that increasing budget support could be a possible response by traditional donors to competitive pressure from China. However, many donors have become more wary of budget support in the years since the global financial crisis, as aid budgets were reduced and recipient countries made less progress in achieving anticipated reforms than expected (Swedlund 2017a).

and development finance can be attractive for governments to claim credit for providing public services and development goods (Cruz and Schneider 2016; Dietrich and Winters 2015). The emphasis on large public infrastructure projects in Chinese development finance makes it particularly attractive for credit-claiming by recipient governments. As Swedlund (2017b: 404-405) points out, the process of disbursing Chinese finance, which tends to be rapid and streamlined, is particularly suited to meeting political demands to demonstrate development through investment in infrastructure.

Financing for infrastructure is also attractive to recipient countries for the development model that it supports. China's own emphasis on a state-led infrastructure-intensive growth trajectory has inspired many governments in developing countries to pursue similar approaches. Part of the success of Chinese development finance in Ethiopia, for instance, has been connected to Ethiopian political elites' attraction to a Chinese model of development that emphasizes public investment in sectors intended to boost exports (Fourie 2015). Similarly in Zambia, early political enthusiasm about Chinese development finance was associated with interest in a Chinese model of development as an alternative to Western approaches to development (Kragelund 2014).

Even in developing countries with liberal market-led approaches to development, the Chinese emphasis on state-led infrastructure investment has shaped expectations about development interventions. For instance, President Uhuru Kenyatta of Kenya praised the Chinese approach by saying, "The [Chinese] Belt and Road initiative gives our continent the opportunity to make a paradigm shift. Post-colonial Africa has been stuck in a rut" (Ilado 2017). Furthermore, as numerous studies on "infrastructure gaps" attest, financing for infrastructure in developing countries is scarce, making China's provision of this type of development finance particularly attractive (African Development Bank, Chief Economist Unit 2010; Serebrisky et al. 2015). For some countries, a preference for infrastructure-focused development finance also reflects an interest in a development model premised on public investment for infrastructure-focused development finance thus also reflects, for some countries, an interest in a certain kind of development model, one that is premised on public investment in infrastructure.

Of course, Chinese-financed projects can also be attractive to recipient country governments for less developmentally-minded reasons. The Chinese approach to development finance has been associated with increased perceptions and experiences of corruption. For example, Brazys et al. (2017) find that Chinese development projects in Tanzania were associated with higher local experiences of corruption. Across sub-Saharan Africa, Isaksson and Kotsadam (2018) find that local reports of corruption increase in the immediate areas surrounding a Chinese-financed project. Thus, in some cases the attraction of Chinese-style development projects for recipient governments may be driven by narrower self-interested reasons, rather than political or developmental motivations. Traditional donors are likely less willing to adjust their development practice to meet these recipient preferences. Nevertheless, politically- and developmentally-motivated recipient preferences for Chinese finance mean that traditional donors could still face competitive pressure.

Political incentives thus lead recipients to prefer development finance that supports large-scale infrastructure investments, such as the type that Chinese official

finance tends to support. However, developing country governments may not wish *all* of their development finance to be of this type. In fact, it is conceivable that the optimum outcome for recipient countries is a specialization among donors, with traditional donors supporting projects in social sectors, including health, education, and governance, and infrastructure projects financed with Chinese support.

This more varied set of recipient preferences with respect to the focus and type of development projects affects predictions about donors' best responses to competition from China. Much of the existing work on donor competition has focused on those aspects of aid where recipients have a linear utility function, preferring more aid to less (Fuchs et al. 2015; Barthel et al. 2014; Granath 2016), and preferring fewer conditions to more (Hernandez 2017). However, recipients may have more complicated preferences with respect to the *type* of development interventions that are supported. Infrastructure-focused projects are preferred, but not all development interventions need be of this type. While the best response for donors competing on the quantity and conditionality of aid are fairly evident - provide more aid, attach fewer conditions - this is less straightforward when competition affects the *type* of development projects supported.

There are at least two approaches that traditional donors can take when confronted with Chinese competition. They can emulate the Chinese approach to development finance and offer the same type of development projects to retain their access to the government or they can choose to differentiate themselves by specializing in a complementary niche with their development projects. Furthermore, it is possible that traditional donors do not respond to the rise of alternative donors by altering their approach to development finance at all. The World Bank may find it too costly to alter the type of development projects they extend to recipient countries, constrained by bureaucratic structures and shareholder preferences.

2.3 Strategies in response to competition

Emulation may be the most direct strategy for donors to ensure they remain attractive to recipient governments and retain access to recipient policymaking. As they observe Chinese financing winning over recipients with large infrastructure projects, donors may reason that the easiest way to retain "market share" is to offer the same kind of aid.

As a multilateral donor, the World Bank is well placed to respond by emulating the Chinese approach and increasing the funding it dedicates to large, infrastructure-intensive projects. Bilateral donors may have less success with such a strategy. At the same time as China's role as a provider of development finance has been increasing, many donor countries have reduced aid budgets and required their development agencies to specialize to achieve development impact with fewer resources. Though the World Bank is not immune to reductions in aid budgets and donor fatigue, it has a larger pool of resources from which to draw, and its IBRD borrowing window is able to rely on funds raised in international capital markets for its financing operations (Humphrey 2017). This gives it resources for the types of projects that characterize Chinese development finance.

If the World Bank responds to Chinese competition by *emulating* the Chinese approach to development finance, then one would expect to observe that recipient countries receiving more Chinese financing commitments also receive more "Chinese-style" aid and finance from the World Bank in subsequent years, i.e. more projects in infrastructure-intensive sectors.

Yet, there are also reasons the World Bank may prefer to respond to the competitive threat of Chinese financing by instead specializing in sectors that are distinct from the Chinese approach. Emulating China's approach to development projects may not always be possible with the resources available. Despite having access to international capital markets, the World Bank is constrained in its ability to meet developing countries' infrastructure investment gap by limited shareholder capital. As Humphrey (2015) explains, the World Bank and other multilateral development banks are forced to take a more cautious approach to their lending since shareholders are reluctant to provide paid-in capital. A more effective use of the World Bank's lending capacity may therefore be to specialize in smaller projects.

Furthermore, the World Bank's comparative advantage relative to China is likely to lie in policy advice and development projects targeted at "soft" sectors. Since at least the mid-1990s, when World Bank President James Wolfensohn defined the identity of the Bank as the "Knowledge Bank," the World Bank has sought to define its value for recipient countries in terms of its technical expertise (Kramarz and Momani 2013).

If the World Bank responds to Chinese competition by *differentiating* its development assistance, then one would expect to observe that recipient countries receiving greater commitments of Chinese development finance subsequently receive a smaller share of World Bank projects in infrastructure-intensive sectors.

Current internal debates within the World Bank about development policy would provide support for either strategy. On the one hand, there has been an increasing emphasis in development policy circles, including within the World Bank, on the importance of financing for infrastructure investment and economic transformation, in part driven by the example of China's development experience (UNCTAD 2014; Lin and Wang 2017). This focus on economic transformation might provide the justification for greater focus on infrastructure financing in a time of greater diversity and potential competition among development financiers. On the other hand, the World Bank has retained its emphasis on interventions in health and education as a means of achieving its goal of reducing extreme poverty, as well as the new goal of "shared prosperity" introduced in 2013 to mitigate inequality in developing countries (World Bank 2013). Given both plausible sets of incentives, to emulate and to differentiate, it is not self-evident which strategy will predominate in response to China's growing role in development finance.

Finally, it is conceivable that the World Bank does not alter its strategy in response to recipient countries' receipt of Chinese finance. In other words, there could be no relationship between China's development finance commitments and the sectoral focus of the World Bank's development projects. This null result would be indicative of the fact that the World Bank does not experience competitive pressure from

Chinese finance or that bureaucratic constraints impede its ability to alter sectoral allocation in response to competition.

3 Data and methods

The empirical analysis uses panel data on over 100 countries that received World Bank or Chinese development finance, or both, over the time period 2000–2014. The key explanatory variable is taken from the AidData Global Chinese Official Finance dataset and measures logged total financing commitments from China to a given country in a given year (Dreher et al. 2017).[13] In all models, Chinese commitments are lagged by one year to account for the delay in responses to the announcement of Chinese financing commitments.

The dependent variable captures whether the World Bank adjusts its development finance in response to competition from Chinese development finance. It measures the share of World Bank projects in a given year that are in "hard" sectors, such as transport, manufacturing, or telecommunications, as opposed to "soft" sectors, such as health or education. Projects are recorded as being in hard sectors if they fall into one of the following sectors in the OECD's Creditor Reporting System (CRS) sector classification scheme: (1) water supply and sanitation, (2) transport and storage, (3) communications, (4) energy, (4) agriculture, forestry & fishing, and (5) industry, mining, and construction.[14] I then calculate the share of the World Bank's annual development projects that are in these hard sectors and use this as the dependent variable.

If the World Bank emulates China in response to competition, then one would expect to observe a positive association between Chinese commitments and the share of World Bank projects in "hard" sectors. By contrast, if the World Bank differentiates itself in response to competition, then one would expect to observe a negative association between Chinese finance and the share of World Bank projects in infrastructure-intensive sectors.

There are good reasons to believe that the relationship between the sectoral focus of World Bank finance and Chinese commitments is endogenous, though the direction of bias is not obvious ex ante. First, it is plausible that developing countries attracting higher quantities of Chinese official finance are particularly well-suited to infrastructure-intensive projects, whether due to their level of development, natural

[13] A value of one is added to Chinese commitments prior to taking the log to avoid the undefined log of zero.

[14] Internally the World Bank uses a different sector classification scheme than the OECD CRS, but AidData's dataset of World Bank projects has assigned sector codes from the OECD CRS coding scheme to World Bank projects to make these comparable. While OECD members assign their projects to only one sector and the AidData dataset similarly assigns Chinese projects to only one sector, the World Bank assigns its projects to up to five sectors. To make the World Bank's share of hard sector projects comparable, I therefore code a World Bank project as being in a hard sector if the largest sector of a project is a "hard" sector.

resource endowments, or other unobserved attributes of their governance or economy. If both China and the World Bank respond to these underlying attributes of recipient countries, this would lead to an upward bias in the estimate of the relationship between Chinese commitments and the type of projects funded by the World Bank. This is not the only possible direction of bias, however. It is also plausible that China is less likely to provide finance to countries that already receive a large number of infrastructure projects from the World Bank, since China may enjoy less access to these recipient governments or identify less of a need for Chinese-funded projects. This would bias downward the estimates of the relationship between Chinese finance and traditional donor allocation.

Furthermore, there are reasonable concerns about reverse causality that could also bias estimates. It is possible that allocations of Chinese official finance are in fact determined by traditional donors' allocation of infrastructure-intensive projects, driven by Chinese efforts to compete or spillover effects.

To address these counfounders to inference, I draw on an instrumental variable approach that has been applied in other studies of Chinese development finance (Dreher et al. 2017; Bluhm et al. 2018). This approach uses annual Chinese steel production to instrument for Chinese overseas finance. Steel is a major input into projects funded by Chinese overseas finance, and excess steel production is therefore associated with higher levels of development finance commitments. This time-varying "shock" of excess Chinese steel production is interacted with a developing country's probability of receiving Chinese finance (the proportion of years in the sample in which a country receives any Chinese finance) to yield an instrument that varies across countries and over time.

To satisfy the exclusion restriction this interacted instrument must not affect the type of projects extended by traditional donors except through the allocation of Chinese finance, i.e. neither the probability of receiving Chinese finance nor Chinese steel production should directly affect the type of projects allocated by traditional donors. While this is implausible on its own, the inclusion of country- and year-fixed effects control for the direct effect of both the probability of receiving Chinese finance and Chinese steel production. The approach is thus analogous to that of a difference-in-differences regression, estimating the differential effect of Chinese steel production on traditional donor behavior in countries with a high probability of receiving Chinese finance compared to a low probability. This requires that World Bank project allocation in countries with differing probabilities of receiving Chinese finance not be affected *differently* by changes in steel production, other than via the impact of Chinese commitments, controlling for country- and year-fixed effects (Dreher et al. 2017: 13).

The instrument would no longer be exogenous if changes in steel production affected World Bank behavior differentially in those countries with a high probability to receive aid compared to countries with a low probability to receive aid, for reasons unrelated to aid. While Chinese steel production may affect the infrastructure-intensity of traditional donor aid directly, this effect is unlikely to be different for high and low recipients of Chinese finance. For instance, the World Bank may be more willing to take on infrastructure-intensive projects when inputs sourced from China are cheap. However, for the exogeneity of the instrument to be violated in this

case, the direct effects of Chinese steel production on the World Bank's allocation of development projects would have to be *different* for countries with a high and low probability of receiving Chinese finance, which is less likely.

In using a time-varying shock that is interacted with unit-level exposure the instrument is similar to the approach used in the "China shock" literature (Autor et al. 2013; Ballard-Rosa et al. 2017) as well the interacted instruments used in work on aid (Nunn and Qian 2014; Ahmed 2016). As in a difference-in-differences approach, this requires parallel pre-trends among the groups that have a high and low probability of being exposed to the "shock." Following the advice of Christian and Barrett (2017) and the implementation of the interacted instrument in Bluhm et al. (2018) and Dreher et al. (2019), I visually inspect the trends in Chinese finance and types of traditional donor allocations for countries with high and low probability of receiving Chinese finance. These trends are largely parallel (see Fig. A1 in the Online Appendix).[15] Furthermore, I check that there is not a close correspondence between the trend in Chinese steel production and Chinese commitments or traditional donor behavior in high or low probability recipients. The plots in Figure show there is no non-linear trend in any of these groups that follows the pattern of Chinese steel production more than another.

Since the dependent variable ranges between 0 and 1, models are estimated using fractional probit regression, with the instrument implemented using a control function approach. The control function approach allows for the estimation of instrumental variable models with limited dependent variables such as a ratio that ranges from 0 to 1. In essence, it uses the residuals from the first stage to subsequently "control for the endogeneity" of the endogenous regressor in the second stage (Wooldridge 2015: 424). Both stages draw on 1,000 bootstrap replications to obtain valid standard errors (Wooldridge 2010, 2015). I also report results estimated using OLS and 2SLS, since these capture the direction and magnitude of local average treatment effect, with fewer assumptions about the functional form (Angrist and Pischke 2008: 197-198). These models have standard errors clustered at the country level.

Throughout, the models control for the logged total amount of World Bank finance committed to the recipient country in a given year, since it is very likely to be correlated with the share of hard sector projects and may be correlated with Chinese finance commitments.[16] The models thus estimate the effect of competition with China on the sectoral allocation of World Bank projects while holding constant the amount of finance allocated by the World Bank, which is another plausible means for the World Bank to respond to Chinese competition. In this way, the effect of competition on the type of development finance is isolated. Table 1 reports summary statistics for the variables in the main models.

[15] Additional material for this article is available on the website of the Review of International Organizations.

[16] A value of one is added to World Bank commitments prior to taking the log to avoid the undefined log of zero.

Table 1 Summary statistics

	Variable	Mean	Std. Dev.	Min.	Max.	N
Country	WB hard sector share	0.139	0.257	0	1	1646
	Chinese commitments (log)	2.026	2.503	0	10.476	1646
	Interacted instrument	6.518	3.669	0	13.462	1646
	WB commitments (log)	3.405	2.53	0	8.859	1646
ADM1	WB hard sector share	0.049	0.202	0	1	39914
	Chinese commitments (log)	0.085	0.583	0	9.920	39914
	Interacted instrument	0.39	0.962	0	9.872	39914
	WB commitments (log)	0.672	1.278	0	8.521	39914
ADM2	WB hard sector share	0.012	0.106	0	1	508606
	Chinese commitments (log)	0.01	0.192	0	9.920	508606
	Interacted instrument	0.044	0.293	0	8.974	508606
	WB commitments (log)	0.092	0.466	0	8.521	508606

The models using the control function are estimated as follows.

First stage:

$$\begin{aligned}\text{ChineseFinance}_{it} &= \pi(\text{ChineseSteel}_{t-3} \times \text{AidProbability}_i) \\ &\quad + \delta\text{WBCommitment}_{it} + \alpha_i + \phi_t \end{aligned} \tag{1}$$

Second stage:

$$\begin{aligned}\text{WBHardSectorShare}_{it} &= \gamma\text{ChineseFinance}_{it-1} + \rho\upsilon_{it} \\ &\quad + \delta\text{WBCommitment}_{it} + \alpha_i + \phi_t \end{aligned} \tag{2}$$

In the first stage, π is the coefficient on the interacted instrument, δ is the coefficient on the control for total World Bank commitments, and α_i and ϕ_t are a country- and year-fixed effect, respectively. The second stage is then estimated by including υ_{it}, the residuals from the first stage. ρ acts as a test of the null of exogeneity. A statistically significant value of ρ indicates that Chinese finance is endogenous to the share of World Bank projects in hard sectors. The main coefficient of interest in the second stage is γ, the estimated effect of Chinese finance on the share of World Bank projects in hard sectors. Again, the second stage controls for World Bank commitments and includes country- and year-fixed effects.

Finally, I investigate the robustness of the main result by extending the analysis to the subnational level. This is a harder test of the theoretical expectations about emulation or differentiation than data at the country-year level. A positive relationship between Chinese financing commitments and the sectoral allocation of World Bank development finance at the national level might obscure subnational heterogeneity, with Chinese commitments to district A being met with a response by the World Bank in district B, for example. However, if the mechanism of competition operates

at a subnational level as well, this suggests that competition is playing out in a localized fashion, with donors perhaps competing for access to particularly competent or attractive regions.

Geotagged data on the location of both World Bank and Chinese projects collected by AidData allow me to identify the amount of Chinese finance and sectoral focus of World Bank projects at the subnational level (AidData 2017; Bluhm et al. 2018). The World Bank dataset includes information on 49,104 projects over the period 2000-2014, while the China dataset includes information on 4,518 projects over the same time period. The geolocation of these projects allows the calculation of Chinese finance commitments and the sectoral composition of World Bank commitments at the larger ADM1 administrative level, as well as the smaller ADM2 level.[17] This enables me to extend the analysis to the subnational level, adopting the same specification as at the recipient country-level, instrumenting for Chinese commitments using the interacted instrument and controlling for World Bank commitments. In this case, the instrument is calculated using the probability of receiving Chinese finance at the subnational level, i.e. the share of years in which the subnational unit received any Chinese financing commitments.[18] These models include unit-level fixed effects and country-year fixed effects and have standard errors clustered at the country level.

4 Results

Across specifications, the World Bank consistently allocates a *greater* share of its projects to infrastructure-intensive sectors when a recipient country has received greater amounts of Chinese finance. I interpret this positive relationship to indicate that *emulation* is a more prominent strategy in response to competition with Chinese development finance than differentiation.

The main results are reported in Table 2. I find a positive association between total Chinese finance commitments and the share of World Bank projects in infrastructure-intensive projects across a range of specifications. Columns 1 and 2 first report the uninstrumented relationship between Chinese finance and the share of World Bank projects in hard sectors, estimated using a factional probit and OLS in turn. There is a positive relationship between the amount of Chinese financing commitments and the share of World Bank projects in the following year, though this is not statistically significant when estimated using fractional probit. However, these results cannot be interpreted causally, given the likely sources of endogeneity outlined above.

We therefore turn next to the results of the instrumental variable approach. Column 3 reports the results of the first stage. The interacted instrument is a good

[17]Including all units in countries that have either a Chinese or World Bank project or both in at least one year 2000-2014 leaves 2,851 ADM1 units and 36,329 ADM2 units.

[18]Note that with the much larger number of units, the median probability of receiving Chinese finance at the ADM1 and ADM2 level is zero. The plots in Fig. A2 for the subnational units are therefore calculated by splitting units into the bottom 90% and top 10% probability of receiving Chinese finance, unlike the country-level parallel trends plots, which divide observations by the median probability of receiving Chinese finance.

Table 2 Share of World Bank projects in infrastructure-intensive hard sectors, 2000–2014

	(1) Probit	(2) OLS	(3) First stage	(4) CF-Second	(5) 2SLS-Second
Chinese commitments (log)	0.027	0.007**		0.278**	0.081***
	(0.017)	(0.003)		(0.137)	(0.025)
Instrument			1.509***		
			(0.216)		
Residuals				−0.254*	
				(0.141)	
WB commitments (log)	0.339***	0.043***	−0.007	0.339***	0.042***
	(0.029)	(0.004)	(0.034)	(0.029)	(0.005)
Constant	−2.988***	−0.070***	−1.348***	−2.704***	−0.047
	(0.193)	(0.021)	(0.294)	(0.478)	(0.060)
Observations	1,646	1,646	1,646	1,646	1,646
F-stat			30.828		
Country FE	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	Yes
Nr. of bootstraps			1000	1000	

Robust standard errors in parentheses
*** p<0.01, ** p<0.05, * p<0.1

predictor of Chinese finance commitments to recipient countries, and the first stage F-statistic of 30.828 clears the threshold for a strong instrument. Columns 4 and 5 report a positive and statistically significant relationship between Chinese financing commitments and the share of World Bank projects in hard sectors. Greater amounts of Chinese financing are associated with a greater share of World Bank projects in hard sectors in the following year. This is true regardless of whether the instrument is implemented using the control function approach or 2SLS.[19] Note that in both cases, whether estimated using the control function or 2SLS, the coefficient on Chinese finance is larger in the second stage (columsn 4 and 5) than in the models with the endogenous regressor (columns 1 and 2). This is likely due to the local average treatment effect estimated in the instrumented models being larger than the average treatment effect estimated in first two models. The interacted instrument based on steel production is most likely to affect China's infrastructure-intensive commitments, which in turn likely have a stronger effect on the World Bank's allocation of projects to infrastructure-intensive sectors.

[19]The significant coefficient on the residuals in the second stage of the control function in column 4 indicates that we can reject the null of exogeneity. Chinese financing commitments are endogenous to the share of World Bank projects allocated to infrastructure-intensive sectors.

Substantively, the results of the model in column 4 indicate that moving from one standard deviation below to one standard deviation above the mean of logged Chinese finance commitments is associated with an increase in the share of World Bank projects in hard sectors from 7.6% to 25.8%. This difference in predicted shares is significant at the 90% level. The marginal effects are reported in the Online Appendix in Fig. A3.[20] Note that these effect sizes are calculated using the within variance in Chinese financing commitments, rather than the overall variance, since the fixed effects in the model reduce the variance used to estimate the relationship. The predictions from the model estimated using 2SLS are broadly similar. Here the move from one standard deviation below the mean to one standard deviation above is associated with an increase from 0% to 30.9% in the share of World Bank projects in hard sectors.

These effects are substantively meaningful, given that the average share of annual World Bank projects in hard sectors is 13%. The World Bank's response to Chinese financing commitments constitutes a significant change in the sectoral composition of World Bank projects. Given the cost of infrastructure-intensive projects, this can have a big impact for recipient countries. The average cost of a World Bank project in a hard sector is $116 million, while projects in other sectors have an average cost of $93 million, a statistically significant difference in means. Even a small increase in the share of projects in infrastructure-intensive sectors can therefore translate into more funding.

I investigate the temporal sensitivity of these results in Table A1 in the Online Appendix. There is a positive relationship between Chinese commitments and the share of World Bank projects in hard sectors across lag structures, but this effect is only statistically significant when Chinese finance commitments are lagged by one year. This suggests that responses to competition happen in a short time frame, rather than impacting World Bank programming far into the future. Given the structure of World Bank project approval, which follows a rigid (and usually lengthy) process (Humphrey 2015), future work may wish to identify at which stage of project design and review considerations about competition and alternative finance enter into donors' decision making. The placebo test with a *t+1* lead of Chinese finance has a null effect on the share of World Bank projects in infrastructure-intensive hard sectors, providing reassurance that the *t-1* lag is not picking up a spurious relationship.

With respect to the control for World Bank total commitments, across the four models predicting the share of projects in infrastructure-intensive sectors, total World Bank commitments are a positive and highly statistically significant predictor. As expected, since infrastructure-intensive projects are more expensive, there is a positive association between the amount of funds committed by the World Bank and the sectoral focus of World Bank projects in a recipient country.[21]

[20] Mummolo and Peterson (2018)

[21] In a robustness check, logged population is included as a control, since larger countries may be more likely to receive both Chinese financing and World Bank infrastructure projects. Given the high multicollinearity between population and World Bank commitments I choose not to include both controls in the same model. The positive effect of Chinese commitments on the share of World Bank projects in hard sectors is unchanged when controlling for population and population is not a significant predictor of the share of World Bank projects in infrastructure-intensive sectors.

The interpretation of these main results is that Chinese development finance places competitive pressure on the *type* of development finance provided by the World Bank, incentivizing the Bank to provide more infrastructure funding to keep up with the focus of Chinese finance on infrastructure-intensive sectors. To validate this interpretation of the results, I re-estimate the models with just Chinese financing commitments in hard sectors, i.e. separating out that portion of Chinese finance that most directly triggers competition over the sectoral focus of development finance. These results are reported in the Online Appendix in Table A2. As expected, there is a positive and statistically significant relationship between Chinese hard sector commitments and the share of World Bank projects in hard sectors, including when Chinese hard sector commitments are instrumented using the interacted instrument. This finding indicates that the positive relationship found in the main results is driven by Chinese hard sector commitments.

To directly test whether it is hard sector spending that triggers a competitive response on the part of traditional donors, I further estimate models that include both Chinese hard and soft sector commitments (which together sum to the total annual Chinese finance commitments) in the same model.[22] These models, reported in the Online Appendix in Table A3, show a statistically significant positive relationship for hard sector commitments, but no such significant relationship for soft sector competitors. In this case, the instrument cannot be deployed, since both forms of Chinese finance cannot be instrumented for using the same interacted instrument. Therefore the relationship cannot be interpreted causally, but can be taken as indicative that Chinese hard sector commitments are driving the main results. This is consistent with the argument that the particular focus of finance provided by China places competitive pressure on the World Bank.

The next step in the analysis is to evaluate the same relationship between Chinese financing commitments and the share of World Bank projects in infrastructure-intensive sectors at the subnational level. As mentioned previously, this is a harder test of the theory, requiring competitive dynamics not simply to aggregate at the national level but rather to unfold at the level of individual districts. A positive relationship between Chinese commitments and the sectoral allocation of World Bank development finance would indicate that competition and emulation is taking place not only in the aggregate at the level of the country-year, but also in a more localized fashion.

At both administrative levels for which there is data, I find a positive and significant relationship between Chinese commitments and the share of World Bank projects in infrastructure-intensive sectors, with Chinese commitments exogenized using the interacted instrument. Table 3 reports these results. The results provide evidence that emulation is occurring at the subnational level and the result at the national level is not simply a function of aggregation across districts.

Again, the models control for the amount of World Bank commitments to account for the fact that infrastructure-intensive projects tend to be more expensive. As before, the models are first estimated with the endogenous variable of Chinese commitments

[22] I am grateful to one of the anonymous reviewers for this suggestion.

Table 3 Subnational: Share of World Bank projects in infrastructure-intensive hard sectors, 2000–2014

	ADM1			ADM2		
	(1) OLS	(2) 2SLS-First	(3) 2SLS-Second	(4) OLS	(5) 2SLS-First	(6) 2SLS-Second
Chinese commitments (log)	0.001		0.066**	0.002		0.092***
	(0.002)		(0.029)	(0.001)		(0.025)
Instrument		1.212***			1.680***	
		(0.221)			(0.303)	
WB commitments (log)	0.056***	−0.002	0.057***	0.118***	−0.001	0.117***
	(0.005)	(0.005)	(0.005)	(0.007)	(0.002)	(0.007)
Constant	0.011***	−0.391***		0.001*	−0.065***	
	(0.004)	(0.087)		(0.001)	(0.013)	
Observations	39,914	39,914	39,914	508,606	508,606	508,606
F-stat		30.049			30.681	
ADM1 FE	Yes	Yes	Yes	–	–	–
ADM2 FE	–	–	–	Yes	Yes	Yes
Country-Year FE	Yes	Yes	Yes	Yes	Yes	Yes

Robust standard errors in parentheses
*** p<0.01, ** p<0.05, * p<0.1

and subsequently use the instrumental variable approach to obtain a more plausibly causal estimate. The models include fixed effects for the subnational administrative unit and the country-year. However, the high number of fixed effects means it is not possible to implement this analysis using fractional probit and the control function approach. The models are therefore estimated as linear probability models using OLS and 2SLS, implemented with the *reghdfe* and *ivreghdfe* packages (Correia 2017).

Columns 1-3 report results from the larger ADM1 subnational units, while columns 4-6 results from the smaller ADM2 subnational units. In both cases, there is a positive relationship between the amount of Chinese finance committed to an administrative unit and the subsequent share of World Bank projects in infrastructure-intensive hard sectors. This relationship becomes significant when Chinese finance is exogenized using the interacted instrument. The instrument is a strong predictor of Chinese finance allocations at the subnational level (see columns 2 and 5). These results provide evidence that responses to competition occur at the subnational level, with the share of World Bank infrastructure projects increasing in areas where China commits greater amounts of development finance. This strengthens the case that the World Bank emulates the Chinese approach to development in response to the growing Chinese presence in recipient countries.

5 Conclusion

The arrival of China as a substantial provider of development finance has shaken up the practices of international development. It provides recipient countries with an additional source of development finance, and emphasizes different aspects of development, stressing investments in productive sectors and infrastructure. China's growing role as a provider of development finance introduces competitive pressures for traditional donors, who risk decreased access to recipient governments as China becomes an increasingly dominant player. I suggest that traditional donors can respond to competition from China either by imitating China's approach to development, thereby hoping to remain attractive to recipient governments, or by specializing in a complementary niche that highlights their relevance to recipient governments.

The analysis of the World Bank responses to Chinese finance commitments suggest that emulation is the more common strategy. When countries receive higher commitments of Chinese finance, the World Bank increases the share of its projects in precisely those infrastructure-intensive "hard sectors" that make up the bulk of Chinese finance. Using an instrumental variable strategy that exploits temporal variation in steel production, one of the key inputs into Chinese-financed development projects, I obtain causal estimates of a positive effect of Chinese development finance on the share of World Bank projects in infrastructure-intensive sectors.

The observed pattern of traditional donor emulation in response to country-level Chinese presence is consistent with my argument that traditional donors can react to competition from alternative donors with emulation. However, it is also conceivable that the changes in traditional donor behavior after China has committed finance to a recipient country are driven by non-competitive mechanisms. China's finance commitment could signal that this recipient country is suitable for a particular type of development intervention, i.e. could play an informational role, or could create positive externalities that make it easier for other donors to carry out infrastructure-intensive projects.

Future work should do more to separate out these mechanisms. One way of doing this would be to study whether the same amount of Chinese finance is associated with different patterns of traditional donor behavior depending on whether it is connected to a single project or multiple projects. If Chinese projects are removing bottlenecks for World Bank projects, then single large projects might have a greater effect than multiple smaller projects. Furthermore, future research could do more to identify the conditions under which emulation is the outgrowth of competition and when it is a product of positive externalities. The argument advanced in this paper would suggest that competitive dynamics are more likely when donors particularly value their access to the recipient government or when the divergence in policy preferences between donors is greatest.

This paper contributes to the emerging literature on the interactions between traditional and new donors, highlighting how the World Bank responds to Chinese finance by altering the sectoral focus of its development projects. Future work could examine other dimensions of traditional donor behavior that may change in response to competition from alternative financing sources, including the speed with which projects are approved (building on Kilby 2013), the modality of aid (including budget

support as a potential means of providing flexible aid to compete with the flexibility of alternative finance), and the environmental review procedures that are applied to projects.[23]

This paper finds that the World Bank changes its behavior in response to China's growing presence in developing countries. These findings have considerable implications for the relationship between donors and recipients, as well as for understanding China's impact on the governance and practices of international development. The increasing magnitude of Chinese development finance means that different types of development projects are being financed, providing an alternative to the approach of traditional donors. Dollar (2018) has suggested that increased competition between Chinese policy banks and the World Bank could encourage greater global focus on infrastructure and growth. The results in this paper indicate that China does in fact alter the development practices of traditional donors in recipient countries in a "bottom-up" fashion.

As China continues to assert its growing strength in world politics, currently dominant actors and institutions must consider their responses. In some cases, emulation and convergence may be the best strategies to respond to competition, but in others, specialization and complementarity might leave all actors better off. Evidence from development finance suggests emulation is an emerging strategy adopted by existing institutions. Future work should consider the patterns of response in other issue-areas where emerging economies' influence is growing, such as environmental and social safeguards, export finance, or investment.

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[23] I am grateful to two of the anonymous reviewers for these suggestions of extensions to the finding in this paper.

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